

TEACHING METHOD

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His study aims to compare three teaching methods—traditional, online, and blended (mixed) instruction—to determine whether the method of instruction significantly affects students' posttest scores. Students are randomly assigned to one of the three teaching methods. All students complete the same final examination at the end of the course. Pre-test scores.

Table 1: The number of students for each group

X	id	group	pretest	posttest
1	1	grp1	14.1	14.1
2	2	grp1	14.5	14.3
3	3	grp1	15.7	14.9
4	4	grp1	16.0	15.3
5	5	grp1	16.5	15.7
6	6	grp1	16.9	16.2
7	7	grp1	17.0	16.5
8	8	grp1	17.0	16.6
9	9	grp1	17.3	16.5
10	10	grp1	17.3	16.7
11	11	grp1	17.8	17.3
12	12	grp1	17.9	17.5
13	13	grp1	19.1	19.3
14	14	grp1	19.4	19.0
15	15	grp1	19.8	19.4
16	16	grp2	13.7	12.7
17	17	grp2	14.7	13.1
18	18	grp2	14.9	13.6
19	19	grp2	15.1	13.6
20	20	grp2	15.8	14.2
21	21	grp2	16.4	14.9
22	22	grp2	16.6	16.1
23	23	grp2	16.9	16.1
24	24	grp2	16.9	16.3
25	25	grp2	17.2	15.9
26	26	grp2	17.8	17.4
27	27	grp2	17.8	16.9
28	28	grp2	18.2	17.1
29	29	grp2	18.4	17.3
30	30	grp2	19.3	17.7
31	31	grp3	14.6	11.7
32	32	grp3	15.0	11.9
33	33	grp3	15.5	11.0

X	id	group	pretest	posttest
34	34	grp3	15.7	12.1
35	35	grp3	16.4	12.3
36	36	grp3	16.9	13.6
37	37	grp3	17.1	14.3
38	38	grp3	17.3	14.2
39	39	grp3	17.5	14.4
40	40	grp3	17.6	13.8
41	41	grp3	17.8	14.3
42	42	grp3	17.9	13.8
43	43	grp3	18.4	15.4
44	44	grp3	18.5	15.1
45	45	grp3	19.0	15.5

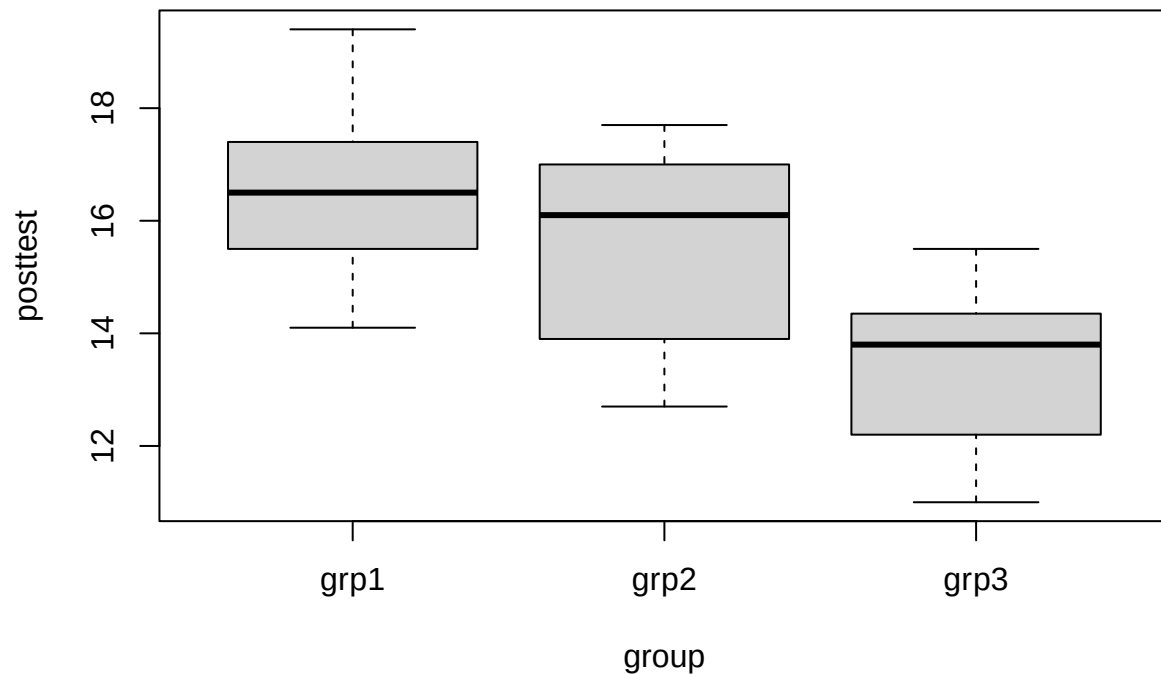
Table one : Shows the number of students in each group and the results of the pre- and post-exams.

Descriptive statistic :

Table 2: Descriptive statistic for students score

group	pretest	posttest
grp1:15	Min. :13.70	Min. :11.00
grp2:15	1st Qu.:15.80	1st Qu.:13.80
grp3:15	Median :17.00	Median :15.30
NA	Mean :16.92	Mean :15.24
NA	3rd Qu.:17.80	3rd Qu.:16.60
NA	Max. :19.80	Max. :19.40

The number of students in the traditional method group is 15, which is equal to the number of students in both the online and blended. methods The average student score on the pre-exam was 16.92, with the highest score being 19.8 and the lowest 13.7. The results show a significant disparity, indicating that student scores are not uniform and are far apart, The average student score on the post-test was 15.24 with the highest score being 19.40 and the lowest 11. The results show that the students' scores were very close and the variation in scores was minimal.



The mean of group1 is greater than mean of group2 and group3 The first group achieved the highest marks compared to the second and third groups.

Fitted model:

```
## The following objects are masked from data2 (pos = 3):
##
##   group, id, posttest, pretest, X

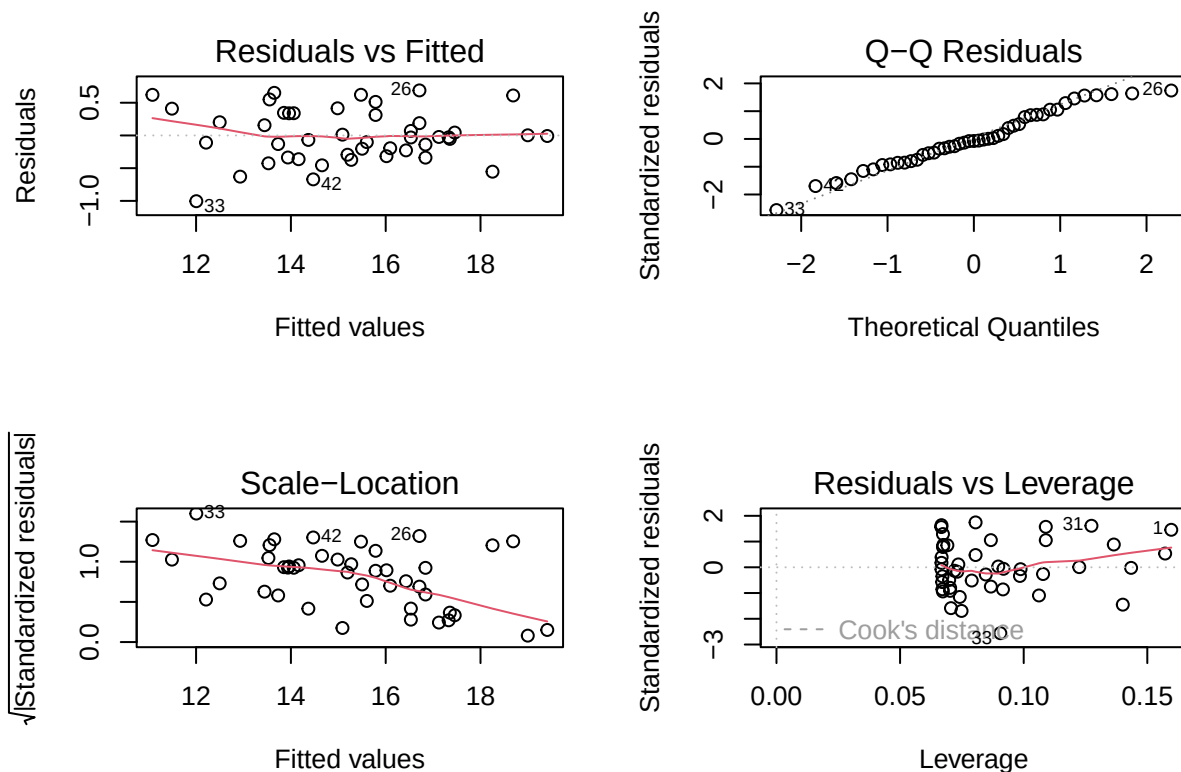
##
## Call:
## lm(formula = posttest ~ pretest + group, data = data2)
##
## Coefficients:
## (Intercept)    pretest  groupgrp2  groupgrp3
##    -0.9409      1.0278    -0.6411    -2.9846

##
## Call:
## lm(formula = posttest ~ pretest + group, data = data2)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.00467 -0.29485 -0.02866  0.33846  0.68799
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept) -0.94088    0.72574   -1.296  0.202074
## pretest      1.02775    0.04202   24.461  < 2e-16 ***
## groupgrp2    -0.64112    0.15137   -4.235  0.000126 ***
## groupgrp3    -2.98463    0.15027  -19.862  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4114 on 41 degrees of freedom
## Multiple R-squared:  0.9615, Adjusted R-squared:  0.9587
## F-statistic: 341.5 on 3 and 41 DF,  p-value: < 2.2e-16
```

The results showed that the pre-test had an impact and relationship with the post-test, where the value reached $p\text{-value} \leq 2e-16$ less than 0.05, The value of $B1^{\wedge}=1.02775$, The value of $B1^{\wedge}$ indicates that for every one increase in the pre-test score, the average score of the post-test will be 1.02775 Ceteris paribus, When transitioning from traditional teaching methods to online learning, the average score on the post-test will be -0.64112, When transitioning from traditional to blended teaching methods, the average score on the post-test equals -2.98463. The value was reached 0.95 This indicates that the model explained 0.95% of the variance in post-test scores, suggesting that the teaching methods and the pre-test were effective.

Diagnostics:



Residuals vs fitted plot showed random scatter, indicating linearity and constant variance assumptions were reasonably met. The Normal Q-Q plot showed points close to the line, normal distribution. Scale-location homogeneous because the points are distributed horizontally. Residuals vs leverage plot showed that the points are inside the shape, no influential points.

##

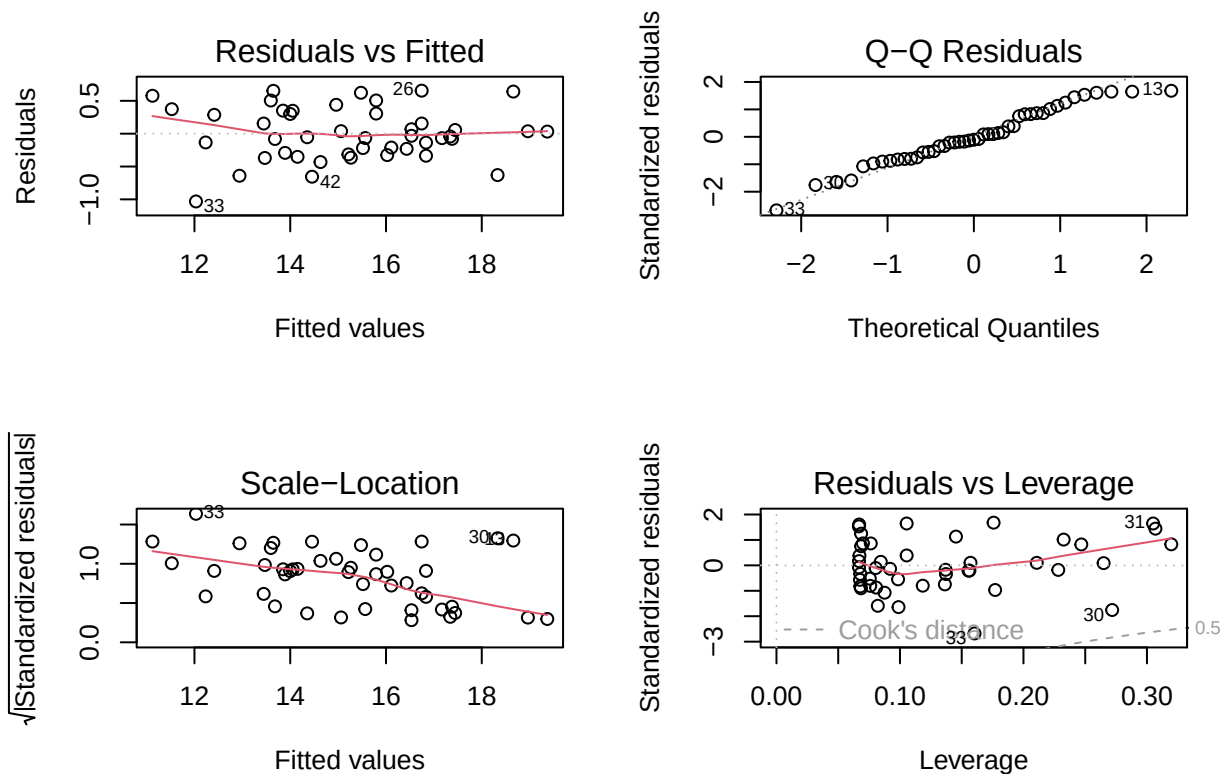
```
## Call:
## lm(formula = posttest ~ pretest + group + pretest:group, data = data2)
##
## Coefficients:
##      (Intercept)      pretest      groupgrp2      groupgrp3
##      -0.685978      1.012835      -1.377018      -2.934908
## pretest:groupgrp2 pretest:groupgrp3
##      0.043813      -0.002987

##
## Call:
## lm(formula = posttest ~ pretest + group + pretest:group, data = data2)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.03176 -0.29238 -0.04249  0.30565  0.65467
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -0.685978   1.184006  -0.579    0.566
## pretest        1.012835   0.069002  14.678 <2e-16 ***
## groupgrp2     -1.377018   1.685672  -0.817    0.419
## groupgrp3     -2.934908   1.872983  -1.567    0.125
## pretest:groupgrp2  0.043813  0.099568   0.440    0.662
## pretest:groupgrp3 -0.002987  0.109531  -0.027    0.978
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4205 on 39 degrees of freedom
## Multiple R-squared:  0.9618, Adjusted R-squared:  0.9569
## F-statistic: 196.2 on 5 and 39 DF,  p-value: < 2.2e-16
```

The model explains 0.9569% of the variance in the dependent variable an exceptionally high explanatory power . Adjusted R-squared=0.9618 The model remains strong even after adjusting for the number of predictors Estimate=1.012,p-value=0.566 < 2e-16 is significant for every one-unit increase in pretest The average score on the post-test is 1.012, all other factors being equal. Estimate=-1.37,p-value=0.419 no statistically significant from the reference group 1 in post test score after controlling for post test When switching from the traditional method to online lower on average decreases by -1.37, all other factors being equal.

Estimate=-2.93,p-value=0.125 no statistically significant from the reference group 1 in post test score after controlling for post test

pretest:group2,Estimate=0.04,p-value=0.662 and group3:pretest,p-value=0.978 no relationship between pretest and post test does not vary by group pretest is the only statistically significant predictor of post test performance .group membership (gro2 and gro3) and interaction do not significant affect post test scores



Residuals vs fitted plot showed random scatter, indicating linearity and constant variance assumptions were reasonably met. The Normal Q-Q plot showed points close to the line, normal distribution. Scale-location homogeneous because the points are distributed horizontally. Residuals vs leverage plot showed that the points are inside the shape, no influential points.

```
## Analysis of Variance Table
##
## Model 1: posttest ~ pretest + group
## Model 2: posttest ~ pretest + group + pretest:group
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1      41 6.9408
## 2      39 6.8961  2   0.04475 0.1265 0.8815
```

The comparison of the two models shows that adding the interaction between pretest and group does not improve the model. The p-value = 0.8815 indicates that the interaction effect is not statistically significant. Therefore, the relationship between pretest and post test does not differ across groups and the simpler model without the interaction is preferred.